

Cybersecurity Incident Report

Section 1: Identify the type of attack that may have caused this network interruption

The logs show repeated SYN requests from 203.0.113.0 to the web server, alongside connection resets and a 504 Gateway Time-out response affecting legitimate clients. This suggests that a SYN flood may be consuming server resources and preventing visitors from accessing the website.

Section 2: Explain how the attack is causing the website to malfunction

1. The client sends a **SYN** packet to the server to request a connection.
2. The server sends a **SYN/ACK** packet back, acknowledging the request and indicating that it is ready to connect.
3. The client sends a final **ACK** packet to acknowledge the server's response. The connection is now established.

In a SYN flood attack, the malicious actor repeatedly sends SYN requests without completing the handshake. The server allocates resources to track these unfinished connections while waiting for the final ACK. This can fill the server's queue of pending connections, preventing legitimate visitors from connecting and causing delays or timeouts.

The logs show repeated SYN requests from 203.0.113.0 to the web server at 192.0.2.1 on port 443. Packets 57, 59 and 61 contain SYN requests, followed by a SYN/ACK reply in packet 64. However, another SYN appears in packet 66 without a final ACK visible in between.

During this traffic, other clients receive connection resets, and one receives an HTTP 504 Gateway Time-out response. These findings are consistent with a suspected SYN flood disrupting access to the website. Further investigation is needed to confirm whether the traffic exhausted the server's resources.